# BowTie Risk Analysis of the Valve Train in a High-Performance VW AP 1.8 Cylinder Head Operating at 10,000 rpm

**Short title:** BowTie risk analysis of the AP 1.8 valve train at 10,000 rpm

**Target journal:** *Reliability Engineering & System Safety* (Elsevier)

**Article type:** Research Paper

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## CRediT authorship contribution statement

**Darlan Costa Porto**: Conceptualization, Methodology, Formal analysis, Investigation, Visualization, Writing – original draft.

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**Florian Pradelle**: Methodology, Validation, Writing – review & editing.

**José Cristiano Pereira**: Conceptualization, Methodology, Supervision, Writing – review & editing, Project administration.

### Declarations

**Competing interests:** The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

**Funding:** This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

**Data availability:** No new experimental data were generated in this study. The Monte Carlo script that reproduces the probabilistic-refinement table (random seed 20260519, $2 \times 10^5$ trials, Python 3.12, NumPy 2.2.6) is provided as supplementary material.